MATHEMATICS

Analysis and approaches

Standard Level | Set B | Paper 1

Mock examination · 2025 series

Time allowed: 1 hour 30 minutes Maximum mark: 80

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

- A calculator is not permitted.
- This paper has Section A and Section B. Both sections are compulsory.
- Answer every question. Show all working and give sufficient mathematical reasoning.
- A clean Mathematics: analysis and approaches formula booklet may be used.
- Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.
- Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.
- The number in square brackets shows the maximum mark for each part.

SECTION A

Question 1 [Maximum mark: 5]

Let $f(x) = 2x - 1$ and $g(x) = 3x$, for $x \in \mathbb{R}$.

(a) Find $(f \circ g)(x)$. [1]

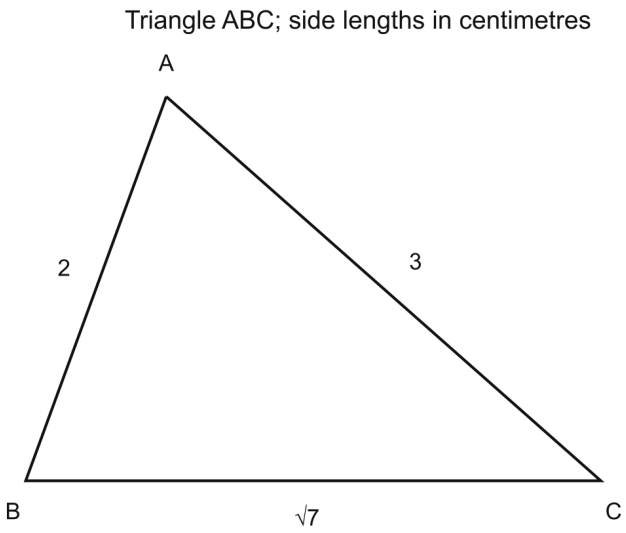
(b) Find $(g \circ f)(x)$ and hence simplify $(f \circ g)(x) - (g \circ f)(x)$. [2]

(c) Find $f^{-1}(x)$. [2]

SECTION A

Question 2 [Maximum mark: 6]

In triangle ABC, $AB = 2\text{cm}$, $AC = 3\text{cm}$ and $BC = \sqrt{7}\text{cm}$.



(a) Find $\angle BAC$ exactly. [4]

(b) Find the exact area of the triangle. [2]

SECTION A

Question 3 [Maximum mark: 5]

Consider the convergent infinite geometric series $4 + 4\left(\frac{a}{7}\right) + 4\left(\frac{a^2}{49}\right) + \cdots$.

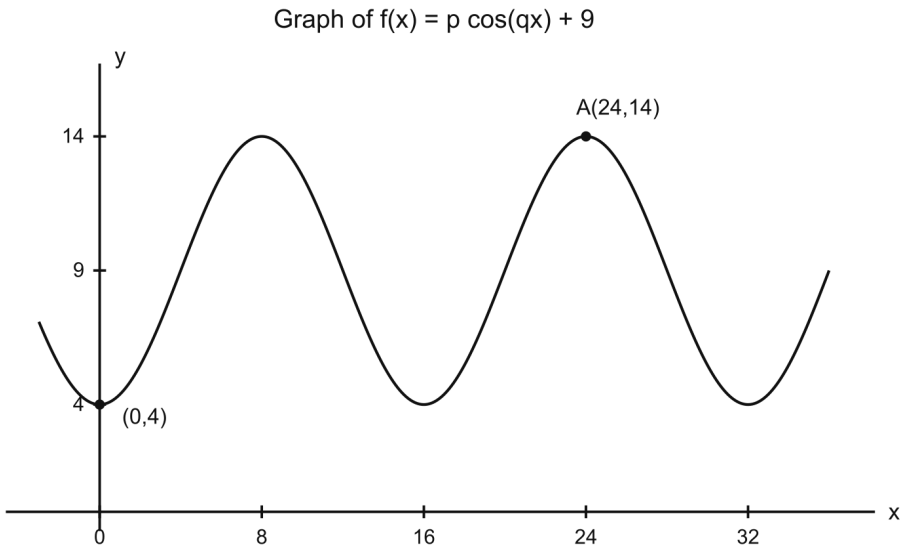
(a) State its common ratio r. [1]

(b) The sum to infinity is 7. Find a. [4]

SECTION A

Question 4 [Maximum mark: 6]

The graph of $f(x) = p\cos(qx) + 9$, where $q > 0$ and x is measured in radians, is shown. It has a local maximum at $A(24, 14)$ and a minimum at the y-intercept $(0, 4)$. The complete cycles visible in the diagram are part of the given information.



(a) Find the period of f. [2]

(b) State the amplitude. [1]

(c) Find p and q. [3]

SECTION A

Question 5 [Maximum mark: 6]

Consider the following integrals.

(a) Find $\int \frac{12x^3}{x^4+1}\,\mathrm{d}x$. [4]

(b) Given $\int_0^1 \frac{12x^3}{x^4+1}\,\mathrm{d}x = \ln A$, find A. [2]

SECTION A

Question 6 [Maximum mark: 6]

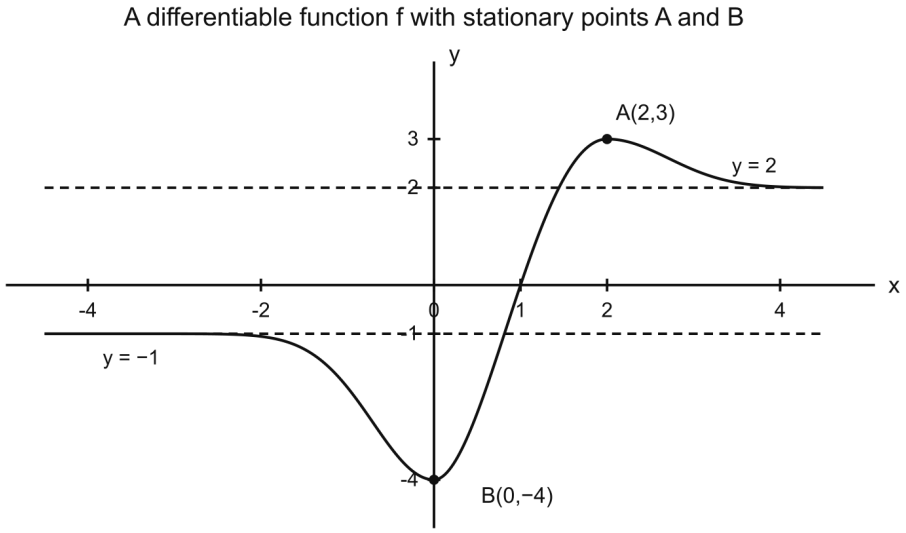
Solve the equation over the real numbers.

$4^{x+1}3^x = 6^{x+1}$. Give an exact answer. [6]

SECTION A

Question 7 [Maximum mark: 6]

The diagram shows a differentiable function f, with stationary points $A(2, 3)$ and $B(0, -4)$. Let $g(x) = x^2 + 1$.



(a) Find $(f \circ g)(1)$ and $(g \circ f)(0)$. [2]

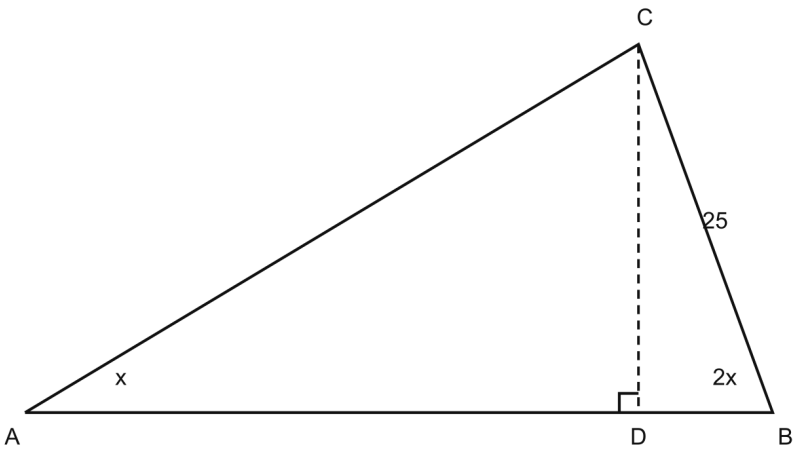
(b) Find the gradient of the normal to $y = f(x)/g(x)$ at $x = 2$. [4]

SECTION B

Question 8 [Maximum mark: 15]

In triangle ABC, $\angle CAB = x$ and $\angle ABC = 2x$, with $0° < x < 90°$ and $\sin x = 3/5$. Side $BC = 25\text{cm}$. Point D is the foot of the perpendicular from C to AB and lies between A and B.

Triangle ABC and perpendicular CD



(a) Find the exact values of $\cos x$ and $\sin 2x$. [4]

(b) Find the exact value of $\cos 2x$. [3]

(c) Find AC. [3]

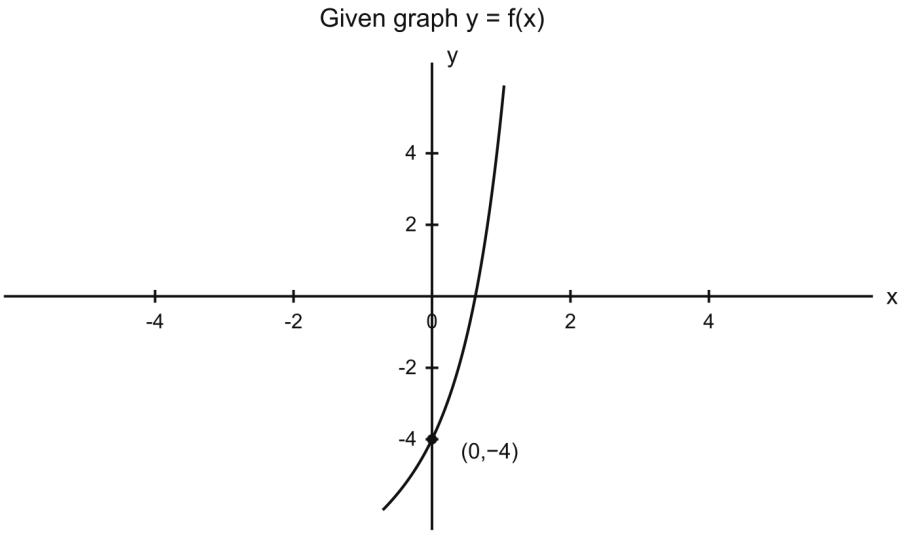
(d) Find BD. [2]

(e) Find the area of triangle ABC. [3]

SECTION B

Question 9 [Maximum mark: 12]

Let $f(x) = e^{2x} + e^x + x - 6$, with domain $\mathbb{R}$. The graph of f is provided.



(a)(i)) Show that f is strictly increasing. [2]

(a)(ii)) Show that its graph is concave upwards. [2]

(b) On the same axes, sketch $y = f^{-1}(x)$ and $y = x$. Label the x-intercept of the inverse and show the common intersection of the three graphs. [4]

(c) The graph $y = 2f(x) + k$ passes through the origin. Find k. [2]

(d) Simplify $f(\ln x) - \ln x$, for $x > 0$. [2]

SECTION B

Question 10 [Maximum mark: 13]

Let $F(u) = (2u - 1)(u - 1)(u + 3)$. All requested solutions are real.

(a) Solve $F(u) = 0$. [2]

(b) Expand $F(u)$ to show that $F(u) = 2u^3 + 3u^2 - 8u + 3$. [1]

(c) Hence solve $2(\ln x)^3 + 3(\ln x)^2 - 8\ln x + 3 = 0$. [3]

(d) Solve $2e^{3x} + 3e^{2x} - 8e^x + 3 = 0$. [3]

(e) Solve $8\sin x - 3\sin^2 x - 2\sin^3 x = 3$ for $0 \leq x \leq 2\pi$. [4]

End of question paper

MATHEMATICS

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Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

Section A

1. [Maximum mark: 5]

$f(x) = 2x - 1, \quad g(x) = 3x$.

(a) $(f \circ g)(x) = 2(3x) - 1 = 6x - 1$. **A1**

[1 mark]

(b) $(g \circ f)(x) = 3(2x - 1) = 6x - 3$ **A1**

$(f \circ g)(x) - (g \circ f)(x) = 6x - 1 - 6x + 3 = 2$. **A1**

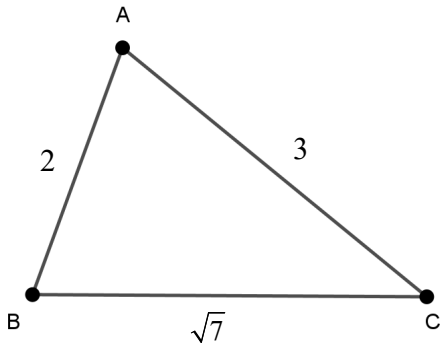
[2 marks]

(c) $f(x) = y \Leftrightarrow 2x - 1 = y \Leftrightarrow x = \frac{y+1}{2}$ **M1**

$f^{-1}(x) = \frac{x+1}{2}$. **A1**

[2 marks]

2. [Maximum mark: 6]



(a) Cosine rule: **M1**

$\sqrt{7}^2 = 2^2 + 3^2 - 2\times 2\times 3\times \cos \hat{BAC}$ **A1**

$\Leftrightarrow 7 = 13 - 12\cos \hat{BAC} \Leftrightarrow 12\cos \hat{BAC} = 6$

$\Leftrightarrow \cos \hat{BAC} = \frac{1}{2}$ **A1**

$\Leftrightarrow \hat{BAC} = 60°$ **A1**

[4 marks]

(b) $A = \frac{1}{2}\times 2\times 3\times \sin 60° = 3\times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$ **M1 A1**

[2 marks]

3. [Maximum mark: 5]

Consider the infinite series $4+4\left(\frac{a}{7}\right)+4\left(\frac{a^2}{49}\right)+\cdots$

(a) $r=\frac{a}{7}$. **A1**

[1 mark]

(b) $S_\infty = 7 \Leftrightarrow \frac{4}{1-\frac{a}{7}} = 7$ **M1 A1**

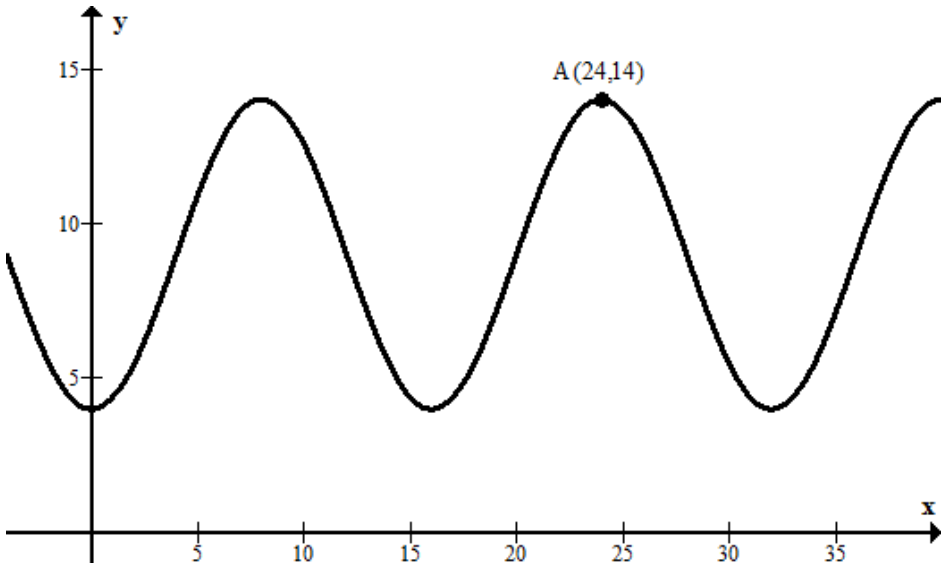
$\Leftrightarrow 4 = 7\left(1-\frac{a}{7}\right) \Leftrightarrow 4 = 7-a$ **A1**

$\Leftrightarrow a = 3$ **A1**

[4 marks]

4. [Maximum mark: 6]

The graph of the function $f(x) = p\cos(qx) + 9$ is shown below



The curve has a local maximum at $A(24,14)$ while the y-intercept is 4.

(a) From 0 to 24: 1.5 periods **M1**

Hence, Period = 16 **A1**

[2 marks]

(b) Amplitude = $14 - 9 = 5$, **A1**

[1 mark]

(c) $p = -5$ (starts from min) **A1**

$$q = \frac{2\pi}{period} = \frac{2\pi}{16} = \frac{\pi}{8}$$ **M1 A1**

[3 marks]

5. [Maximum mark: 6]

(a) $\int \frac{12x^3}{x^4+1}\mathrm{d}x$.

Let $u = x^4+1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 4x^3 \Rightarrow \mathrm{d}x = \frac{\mathrm{d}u}{4x^3}$ **M1 A1**

$\int \frac{12x^3}{x^4+1}\mathrm{d}x = \int \frac{12x^3}{u}\frac{\mathrm{d}u}{4x^3} = \int \frac{3}{u}\mathrm{d}u = 3\ln u + c$ **A1**

$= 3\ln(x^4+1) + c$ **A1**

[4 marks]

(b) $\int_0^1 \frac{12x^3}{x^4+1}\mathrm{d}x = \left[3\ln(x^4+1)\right]_0^1 = 3\ln 2 - 3\ln 1 = 3\ln 2$ **M1**

$= \ln 8$, hence $A = 8$ **A1**

[2 marks]

6. [Maximum mark: 6]

$4^{x+1} \times 3^x = 6^{x+1}$

$\Leftrightarrow 4^x \times 4 \times 3^x = 6^x \times 6$ **M1**

$\Leftrightarrow \frac{4^x \times 3^x}{6^x} = \frac{6}{4}$ **A1**

$\Leftrightarrow \left(\frac{4 \times 3}{6}\right)^x = \frac{3}{2}$ **A1**

$\Leftrightarrow 2^x = \frac{3}{2}$ **A1**

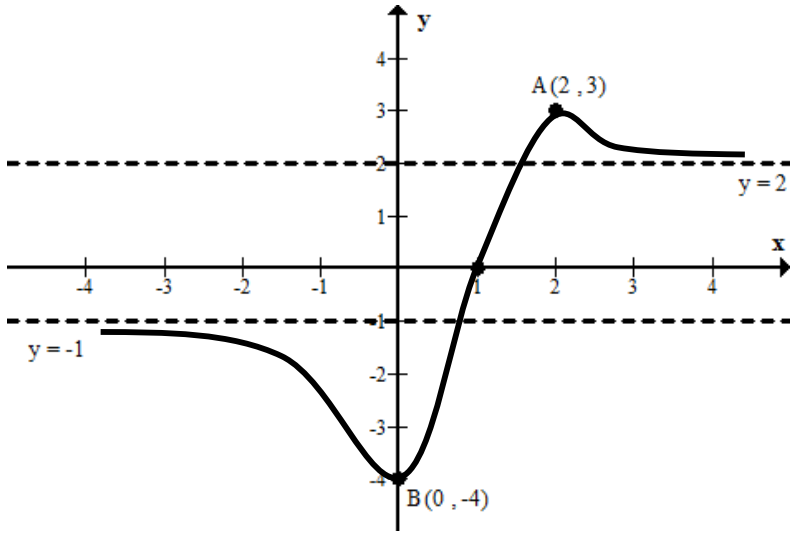
$\Leftrightarrow x = \log_2 \frac{3}{2} = \log_2 3 - 1$ **M1 A1**

[6 marks]

Note: There are several alternative methods, but this is a quick one!

7. [Maximum mark: 6]

$f(x)$



$g(x) = x^2 + 1$.

(a) $(f \circ g)(1) = f(2) = 3$ **A1**

$(g \circ f)(0) = g(-4) = 17$. **A1**

[2 marks]

(b) $\frac{dy}{dx} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$ **M1**

$g'(x) = 2x$ **A1**

$\left.\frac{dy}{dx}\right|_{x=2} = \frac{f'(2)g(2) - f(2)g'(2)}{g(2)^2}$

$= \frac{0 - 3 \times 4}{5^2} = -\frac{12}{25}$ **A1**

$m_N = \frac{25}{12}$ **A1**

[4 marks]

Section B

8. [Maximum mark: 15]

$\sin x = \frac{3}{5},\ 0° < x < 90°$

(a) By using the Pythagorean identity **OR** the triangle method **M1**

$\cos x = \frac{4}{5}$ **A2**

$\sin 2x = 2\sin x \cos x = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}.$ **A1**

[4 marks]

(b) $\cos 2x = 1 - 2\sin^2 x = 1 - 2\left(\frac{3}{5}\right)^2 = 1 - \frac{18}{25} = \frac{7}{25}.$ **M1 A1 A1**

[3 marks]

(c) $\frac{25}{\sin x} = \frac{AC}{\sin 2x}$ **M1**

$\Leftrightarrow AC = \frac{25}{\sin x} \times \sin 2x \Leftrightarrow AC = \frac{25}{\frac{3}{5}} \times \frac{24}{25} = 40$ **A1 A1**

[3 marks]

(d) $\cos 2x = \frac{BD}{BC} \Leftrightarrow \frac{7}{25} = \frac{BD}{25} \Leftrightarrow BD = 7$ **M1 A1**

[2 marks]

(e) $AB = AD + BD = 32 + 7 = 39$ **A1**

Area $= \frac{1}{2} \times 40 \times 39 \times \sin x = 20 \times 39 \times \frac{3}{5} = 468$ **M1 A1**

[3 marks]

9. [Maximum mark: 12]

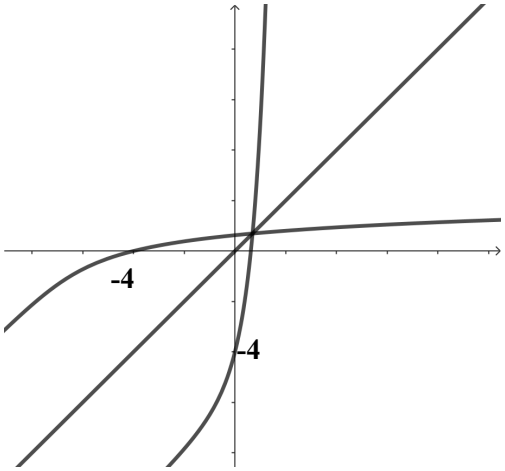
$f(x) = e^{2x} + e^{x} + x - 6$.

(a) (i) $f'(x) = 2e^{2x} + e^{x} + 1 > 0$ for all x, hence increasing. **M1 A1**

(ii) $f''(x) = 4e^{2x} + e^{x} > 0$ for all x, hence concave up. **A1 A1**

[4 marks]

(b)



Inverse, $y = x$ **A1 A1**

Meeting all 3 at the same point **A1**

x-intercept at -4 **A1**

[4 marks]

(c) Using transformations: $(0,-4) \to (0,-8) \to (0,-8+k)$, $k = 8$ **M1 A1**

OR

$y = 2\left(e^{2x} + e^{x} + x - 6\right) + k$, so for $x = 0, y = 0$, $k = 8$ **M1 A1**

[2 marks]

(d) $f(\ln x) - \ln x =$ $e^{2\ln x} + e^{\ln x} + \ln x - 6 - \ln x = x^2 + x - 6$ **M1 A1**

[2 marks]

10. [Maximum mark: 13]

Let $f(u) = (2u-1)(u-1)(u+3)$ **A1 A1**

(a) $u = \frac{1}{2}$, $u = 1$, $u = -3$

[2 marks]

(b) $f(u) = (2u-1)(u-1)(u+3) = (2u^2 - 2u - u + 1)(u+3)$

$= (2u^2 - 3u + 1)(u+3)$

$= 2u^3 + 6u^2 - 3u^2 - 9u + u + 3$. **A1**

$= 2u^3 + 3u^2 - 8u + 3$.

[1 mark]

(c) Solve the equation by letting $u = \ln x$ and using the results in (a) and (b).

$\ln x = \frac{1}{2} \Rightarrow x = e^{\frac{1}{2}} \quad (= \sqrt{e})$ **A1**

$\ln x = 1 \Rightarrow x = e$ **A1**

$\ln x = -3 \Rightarrow x = e^{-3} \quad \left(= \frac{1}{e^3}\right)$ **A1**

[3 marks]

(d) $2e^{3x} + 3e^{2x} - 8e^x + 3 = 0$.

$e^x = \frac{1}{2} \Rightarrow x = \ln \frac{1}{2}$ **A1**

$e^x = 1 \Rightarrow x = 0$ **A1**

$e^x = -3$ (rejected) **A1**

[3 marks]

(e) $8\sin x - 3\sin^2 x - 2\sin^3 x = 3$, for $0 \le x \le 2\pi$.

$\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \quad x = \frac{5\pi}{6}$ **A1 A1**

$\sin x = 1 \Rightarrow \quad x = \frac{\pi}{2}$ **A1**

$\sin x = -3$ (rejected) **A1**

[4 marks]